



Title:

SENTINEL

**Smart Elderly Navigation &
Telehealth Integrated Lifeline**

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Outlines

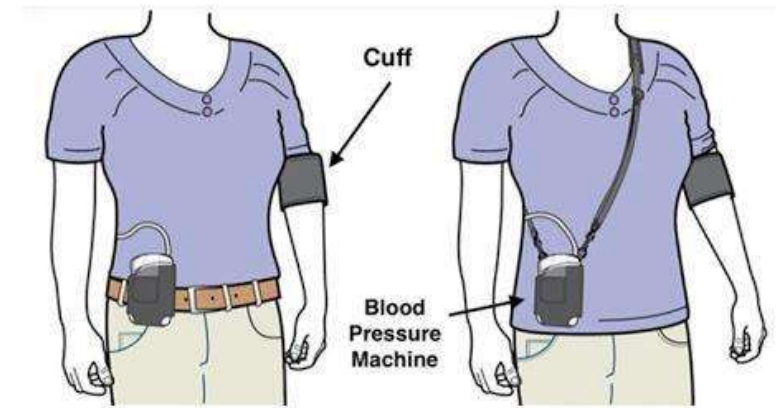


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Introduction

- Cardiovascular diseases cause ~17.9M deaths annually [1]
- Blood pressure is a key indicator.
- Continuous monitoring enables early intervention.
- Cuff devices are inconvenient.
- Cuffless monitoring enables real-time tracking.



Background

Traditional vs Cuffless Blood Pressure Monitoring

Method	Characteristics
ABP (Arterial Blood Pressure)	Invasive
Cuff-based	Non-invasive but intermittent

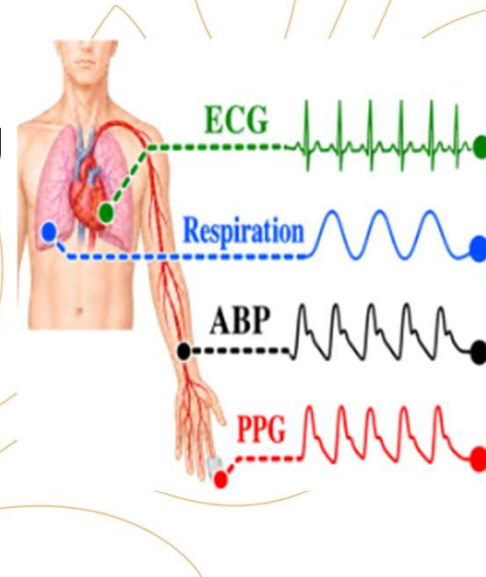
Physiological Signal	Description
ECG	Electrical activity of the heart
PPG	Blood volume changes
RESP	Respiratory activity

Limitations

- No continuous monitoring
- Uncomfortable for frequent use
- Not suitable for wearable devices
- Limited long-term tracking

Advantages

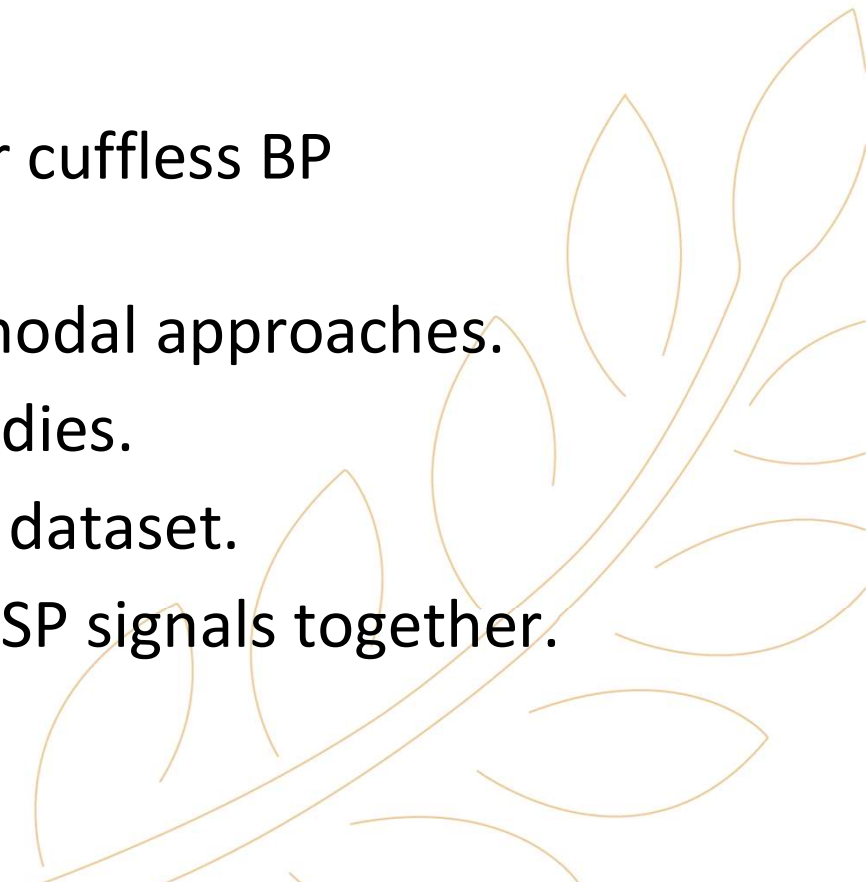
- Continuous monitoring
- Non-invasive
- Wearable-friendly
- Real-time estimation



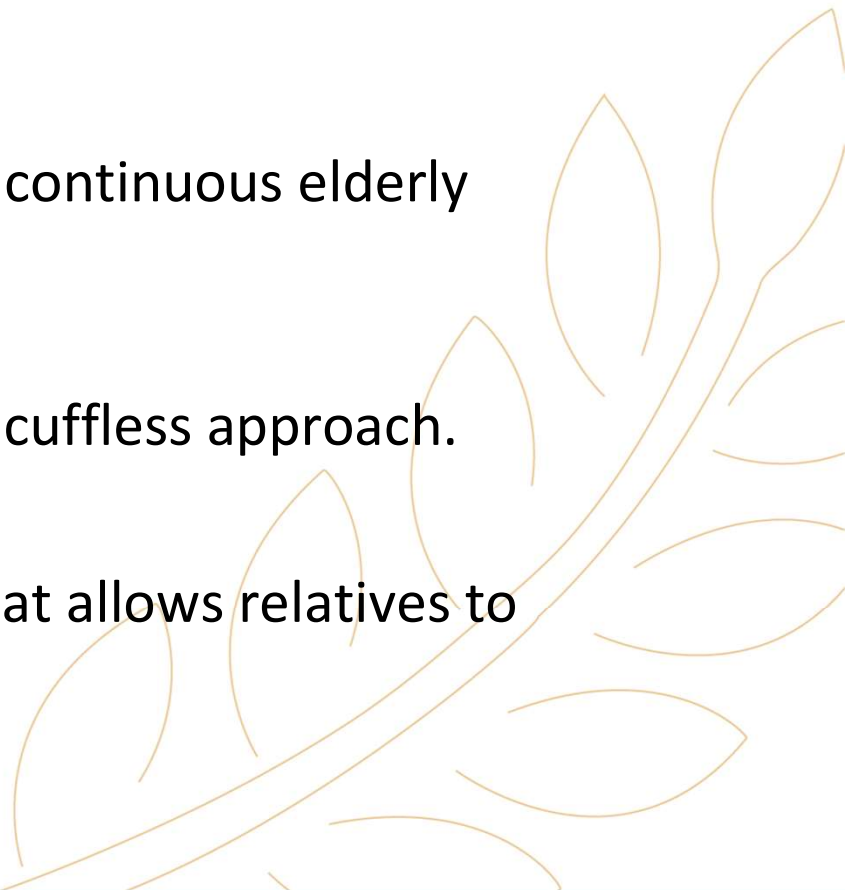
Islam, S. M. S., et al., *Wearable Cuffless Blood Pressure Monitoring Devices: A Systematic Review and Meta-analysis*, European Heart Journal – Digital Health, 2022. ^[2]

Problem Statement



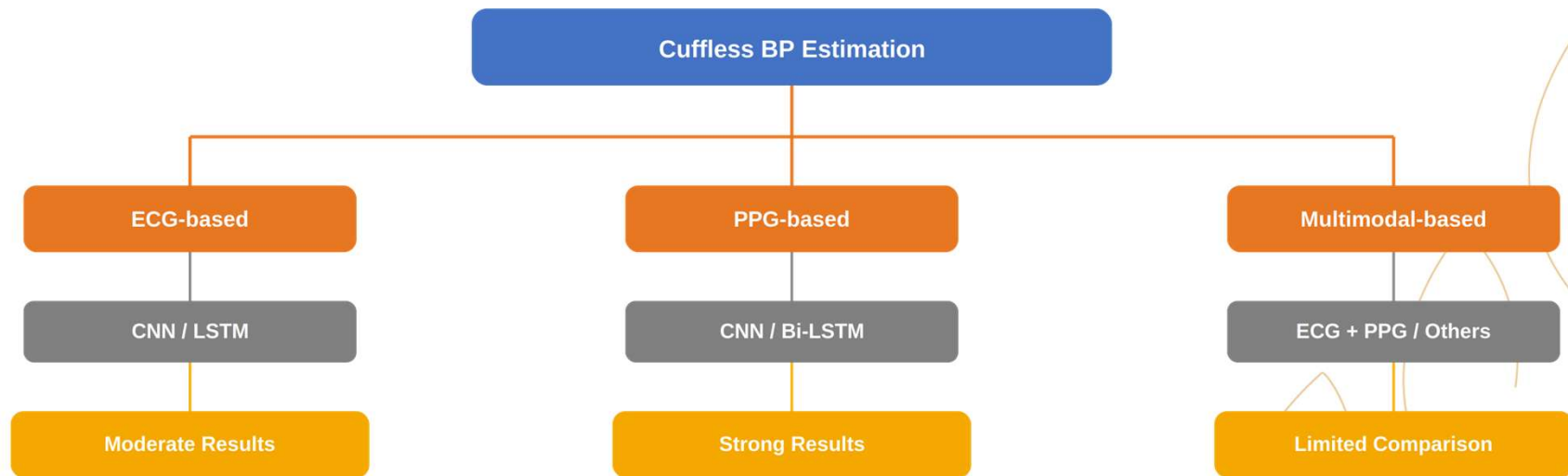
- No clear best physiological modality for cuffless BP estimation.
 - Uncertainty between single and multimodal approaches.
 - Inconsistent results across previous studies.
 - Lack of fair comparison using the same dataset.
 - Limited evaluation of ECG, PPG, and RESP signals together.
- 

Objectives



- Enable real-time blood pressure estimation for continuous elderly health monitoring.
- Replace frequent cuff-based monitoring with a cuffless approach.
- Develop an AI-powered monitoring platform that allows relatives to remotely track patient status and alerts.

Related Work



Conceptual categorization based on the reviewed literature.

Related Work 1/2

Koparı et al. (2024)

- PPG Only
- Model: Deep CNN + Transfer Learning
- CNN achieved strong performance using PPG

Limitation:

- Single modality only
- No architecture comparison
- No multi-task learning

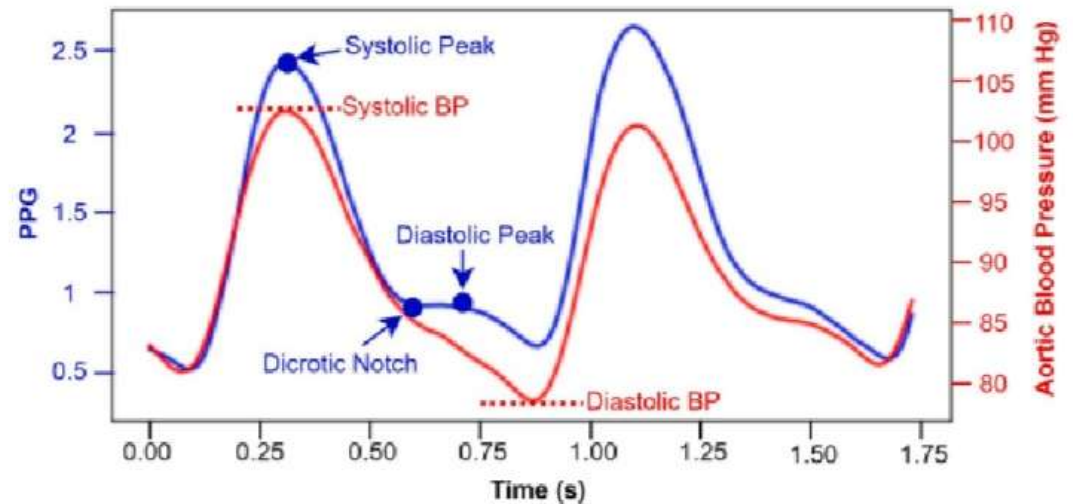


Fig. 2. The waveform and critical points of a typical PPG and ABP signal.

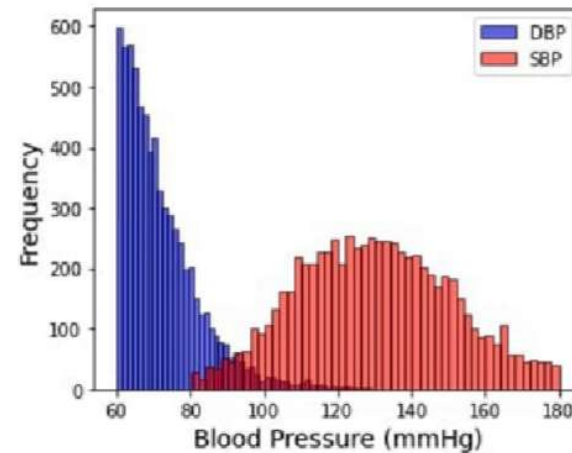


Fig. 3. Distribution of BP values for the final dataset.

Related Work 2/2

Xu et al. (2024)

- ECG + PPG Signals
- Model: Deep Learning Framework
- Combined ECG and PPG improved prediction performance compared to using a single signal.

Limitation:

- ECG & PPG only.
- No RESP signals.
- Limited model comparison

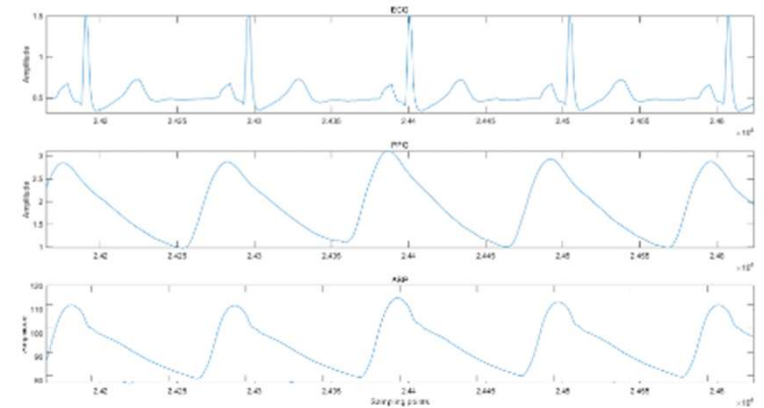


Figure 1: ECG,PPG and ABP signals after data preprocessing.

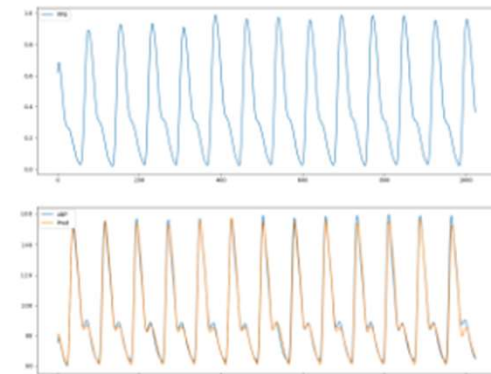


Figure 4: Predicted ABP.

H. Xu, Z. Zhao, H. Wang, L. Zhou, J. Wang, J. Ji, Q. Shi, and S. Zhang, "En-hanced blood pressure estimation based on ppg and ecg signals," in Proceedings of the 2024 International Conference on Smart Healthcare and Wearable Intelligent Devices, SHWID '24, (New York, NY, USA), p. 41–45, Association for Computing Machinery, 2024.[3]

Methods



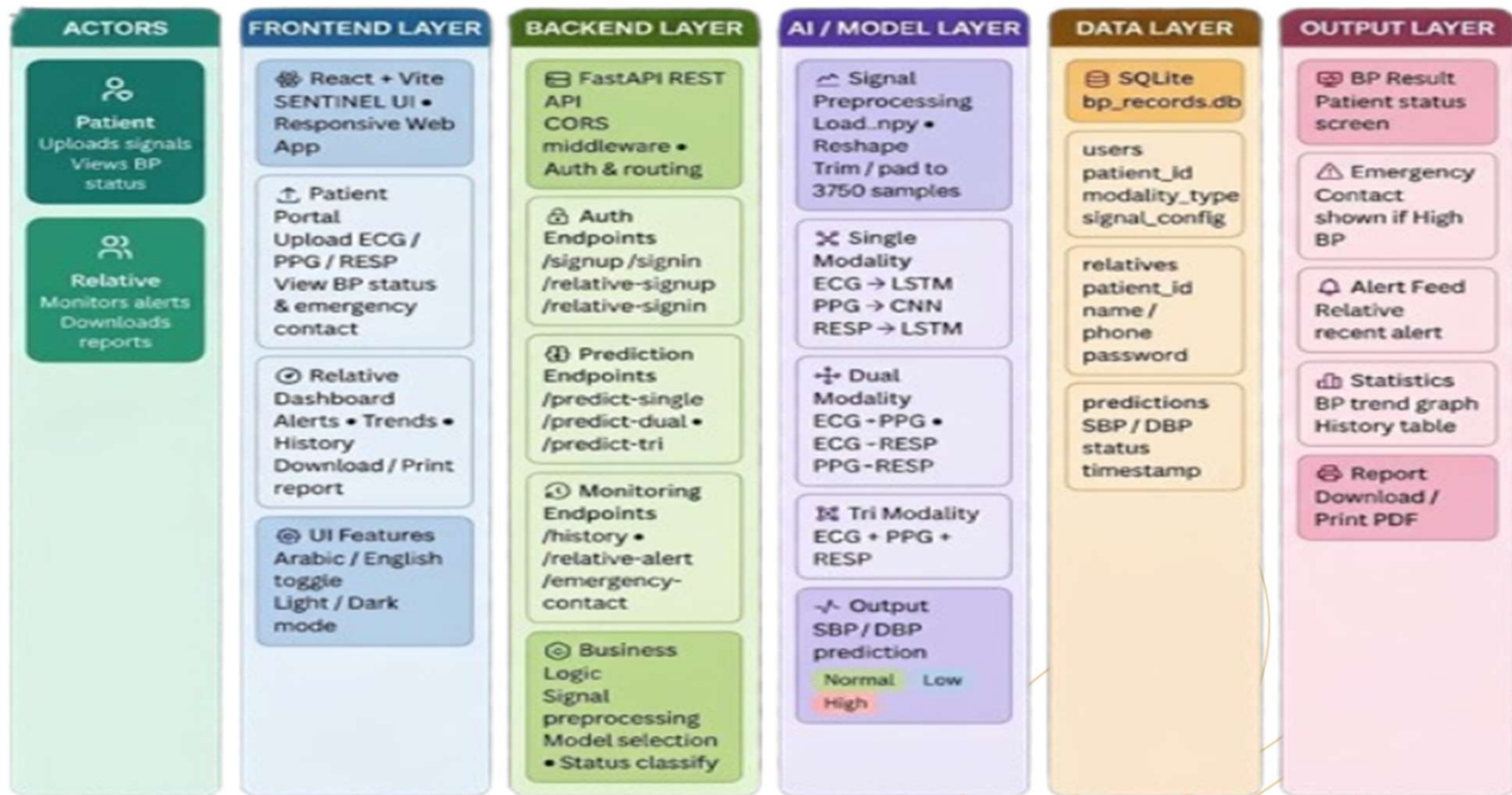
Method 1: Modality Benchmarking

- Compare ECG, PPG, and RESP signals
- Use the same dataset and evaluation protocol
- Identify the best modality for BP estimation

Method 2: Multimodal Learning

- Combine multiple physiological signals
 - Evaluate performance improvement
 - Compare Single, Dual, and Tri modalities
- 

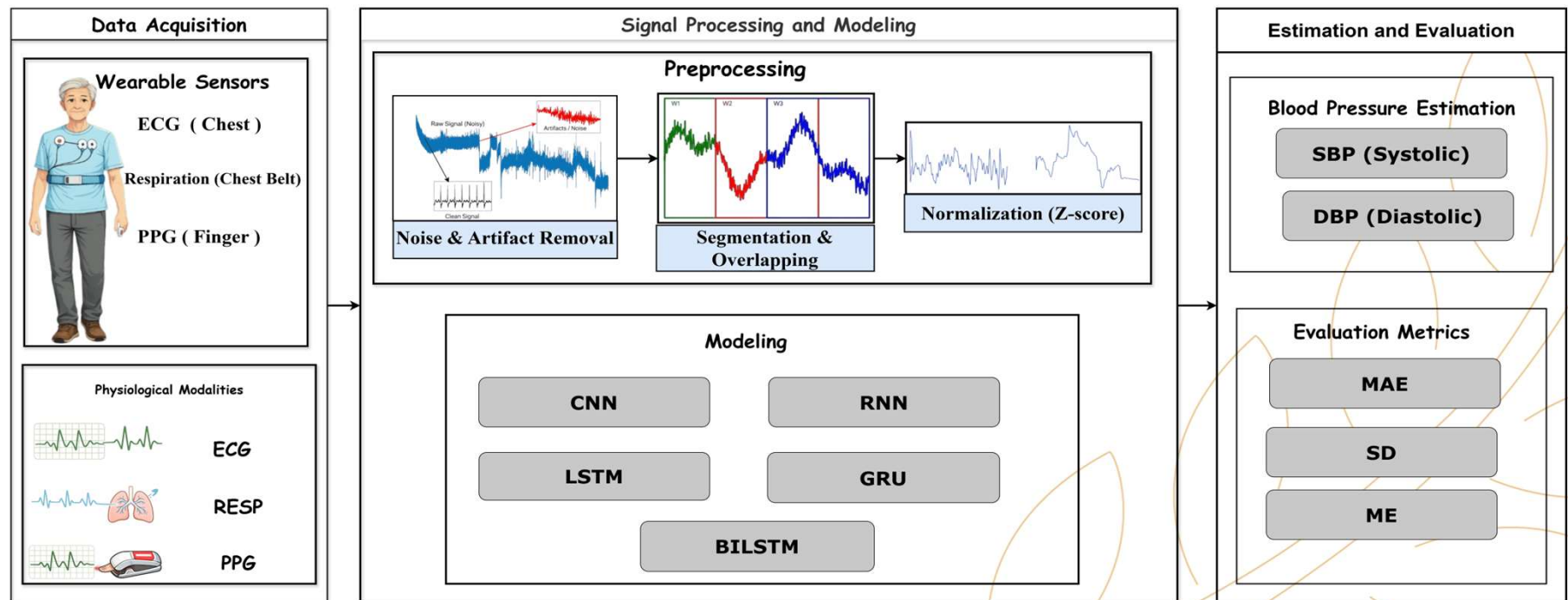
System Architecture



DATA FLOW

Upload ECG/PPG/RESP → React sends to FastAPI → Preprocess signals → Select AI model → Predict SBP/DBP → Classify status → Save to SQLite → Patient & Relative view

Study 1



Experimental Setup



- **Data Split 7:1:2**

Subject-level split

- **Modality specific encoder**

CNN
RNN
LSTM
GRU
Bi-LSTM

- **Training**

Adam Optimizer

Huber Loss

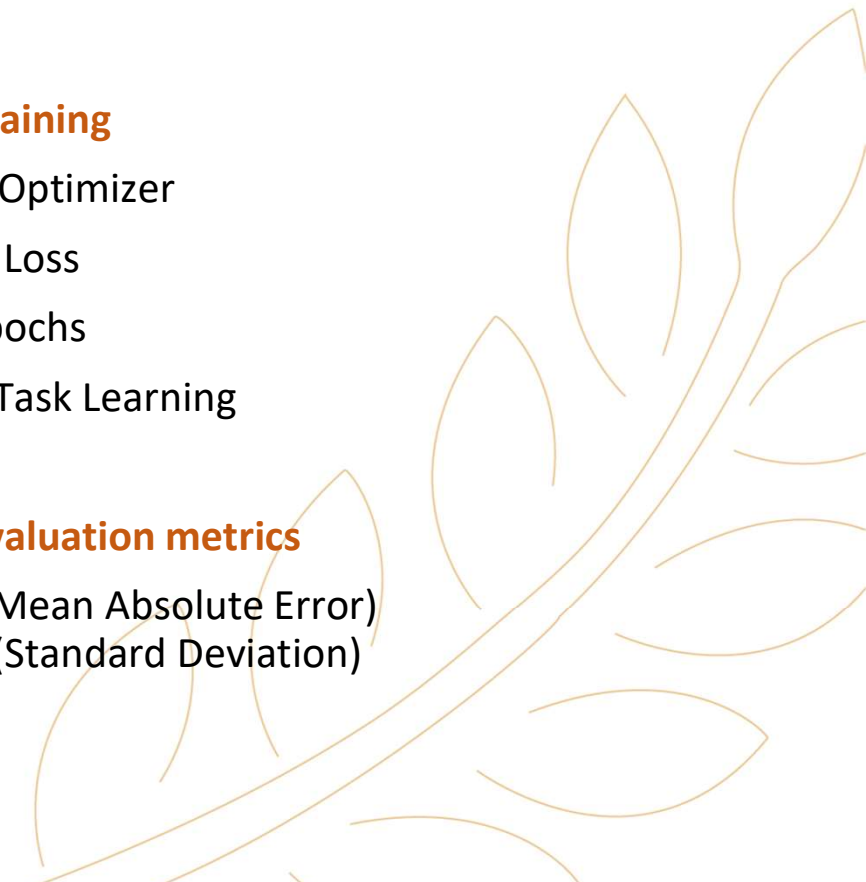
100 Epochs

Multi-Task Learning

- **Evaluation metrics**

MAE (Mean Absolute Error)

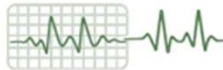
STD (Standard Deviation)



Dataset & Preprocessing

MIMIC-BP Dataset

- 1524 subjects
- ECG signals
- PPG signals
- Respiratory signals (RESP)
- SBP and DBP labels



Why MIMIC-BP?

- Publicly available benchmark
- Large-scale physiological dataset
- Only public dataset containing ECG, PPG, RESP and BP labels

Dataset	ECG	PPG	Resp.	ABP/BP Label
BIDMC [25]	×	✓	✓	×
PulseDB [26]	✓	✓	×	✓
UCI Cuffless BP [27]	✓	✓	×	✓
VitalDB [28]	✓	✓	✓ _{CO₂}	✓
MIMIC-II [29]	✓	✓	✓	✓
MIMIC-III [30]	✓	✓	✓	✓
MIMIC-IV [31]	✓	✓	✓	✓
MIMIC-BP [7]	✓	✓	✓	✓

All modalities follow exactly the same preprocessing pipeline to ensure a fair comparison.

Dataset Analysis

Subjects

1524

Segments

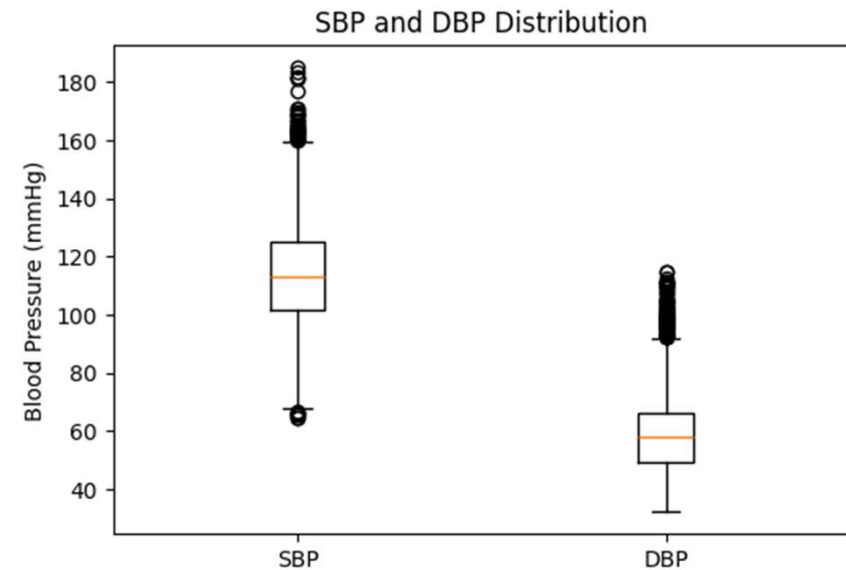
45,720

Window Length

30 sec

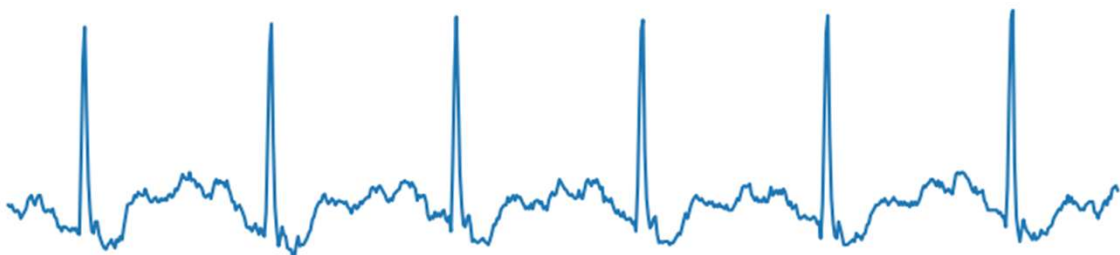
Sampling Rate

125 Hz



SBP values show wider variability than DBP, which makes SBP prediction more challenging.

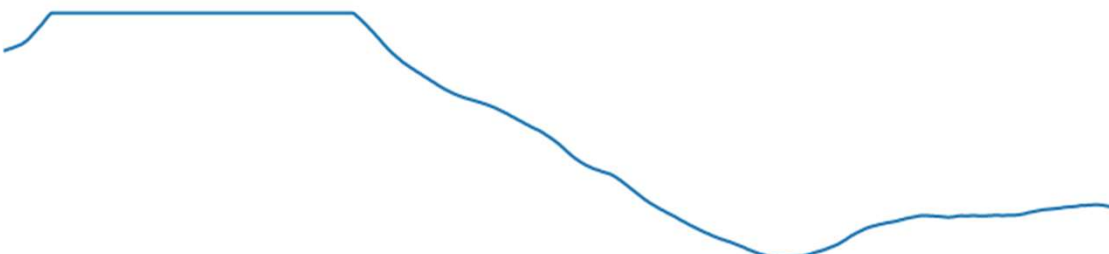
Dataset Samples



ECG Signal



PPG Signal



RESP Signal

Parameter	Value
SBP	113.6 mmHg
DBP	55.6 mmHg

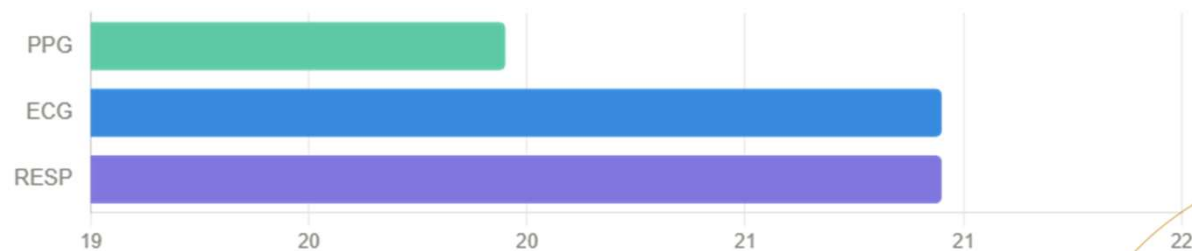
Ground Truth Blood Pressure Labels

Experimental Results (1/2)

Single-Modality Benchmark

Average over 5 independent runs

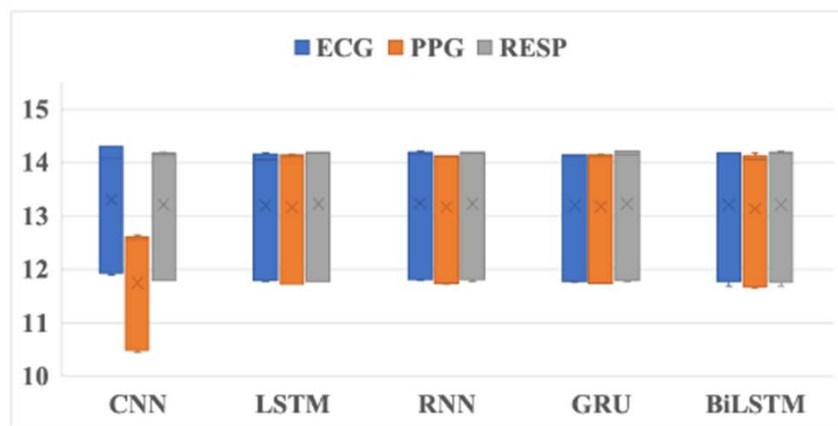
ECG	PPG	RESP
Best Architecture per Modality		
Bi-LSTM	CNN	Bi-LSTM
Avg MAE $\approx$ 22.52	Avg MAE $\approx$ 19.95	Avg MAE $\approx$ 22.55



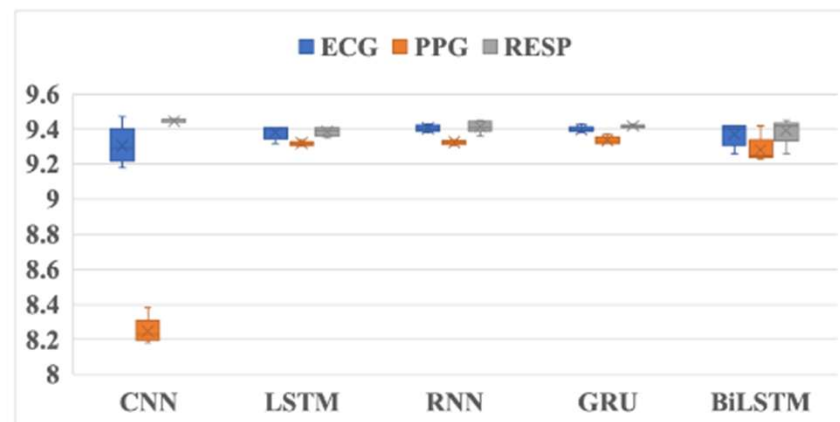
- PPG achieved the lowest MAE (19.95).
- CNN performed best for PPG signals.
- Bi-LSTM performed best for ECG and RESP.
- Modality choice had a larger impact than architecture choice.

Experimental Results (1/2)

Single-Modality Benchmark



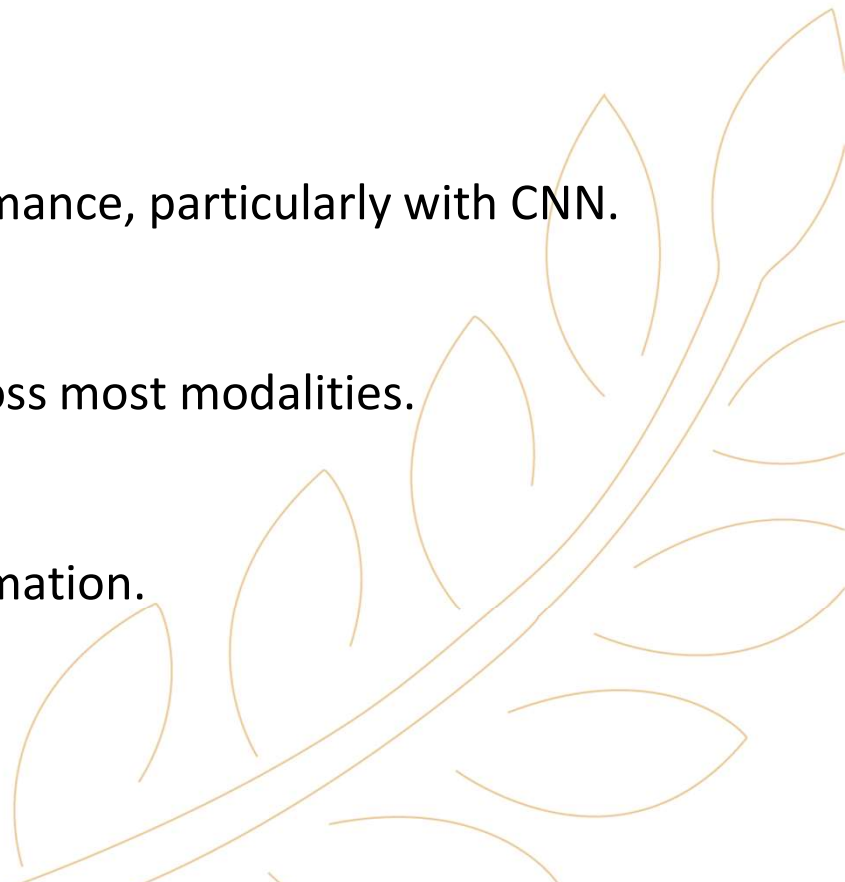
(a) Distribution of SBP estimation errors (MAE) across five runs.



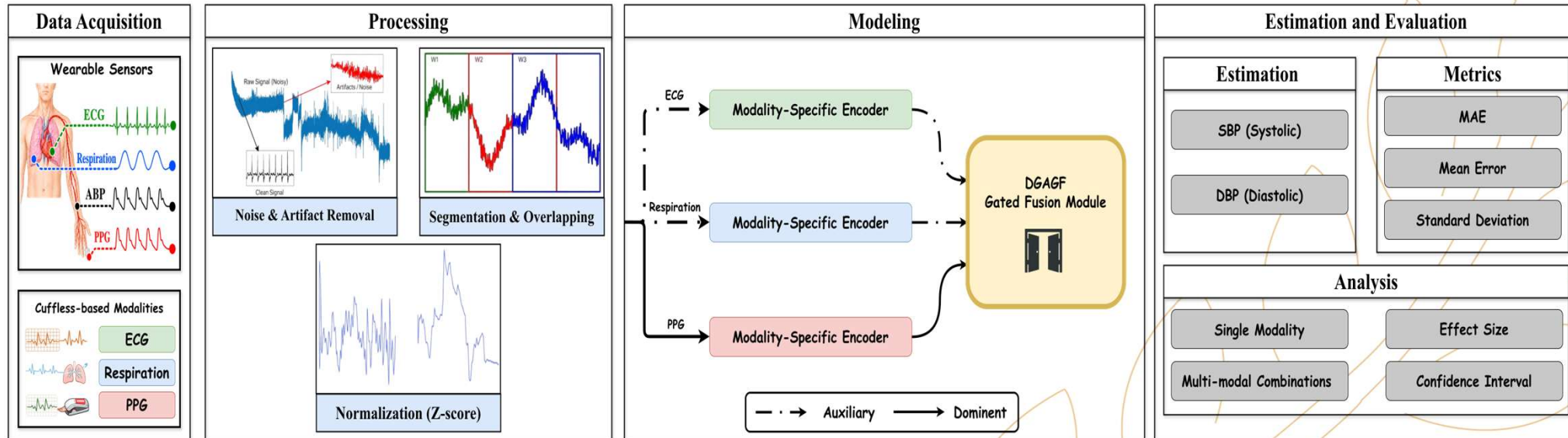
(b) Distribution of DBP estimation errors (MAE) across five runs.

Study 1 findings



- PPG achieved the best overall single-modality performance, particularly with CNN.
 - LSTM and Bi-LSTM achieved similar performance across most modalities.
 - DBP estimation was consistently easier than SBP estimation.
- 

Study 2



Experimental Setup

- **Data Split 7:1:2**

Subject-level split

- **Modality specific encoder**

CNN

Bi-LSTM

- **Fusion mechanisms**

Concatenation

Bidirectional Cross Attention (BCA)

GMU

DGDAF

- **Training**

Adam Optimizer

Huber Loss

100 Epochs

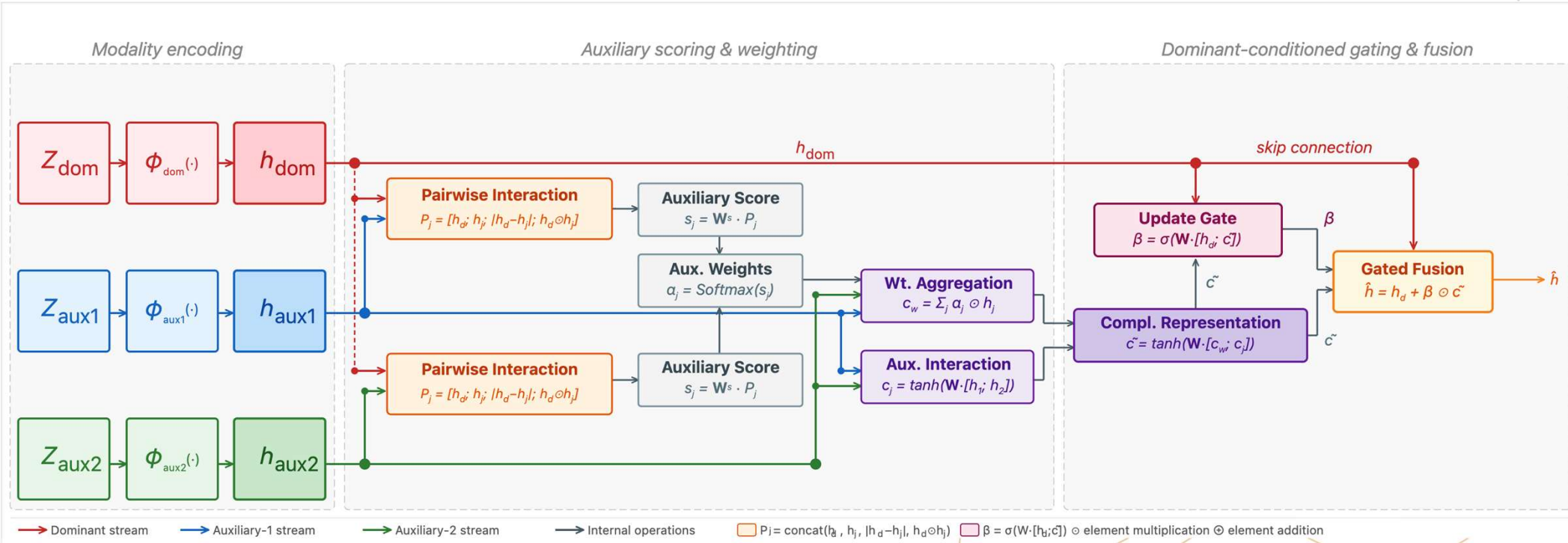
Multi-Task Learning

- **Evaluation metrics**

MAE (Mean Absolute Error)

STD (Standard Deviation)

Architecture of the proposed DGAGF

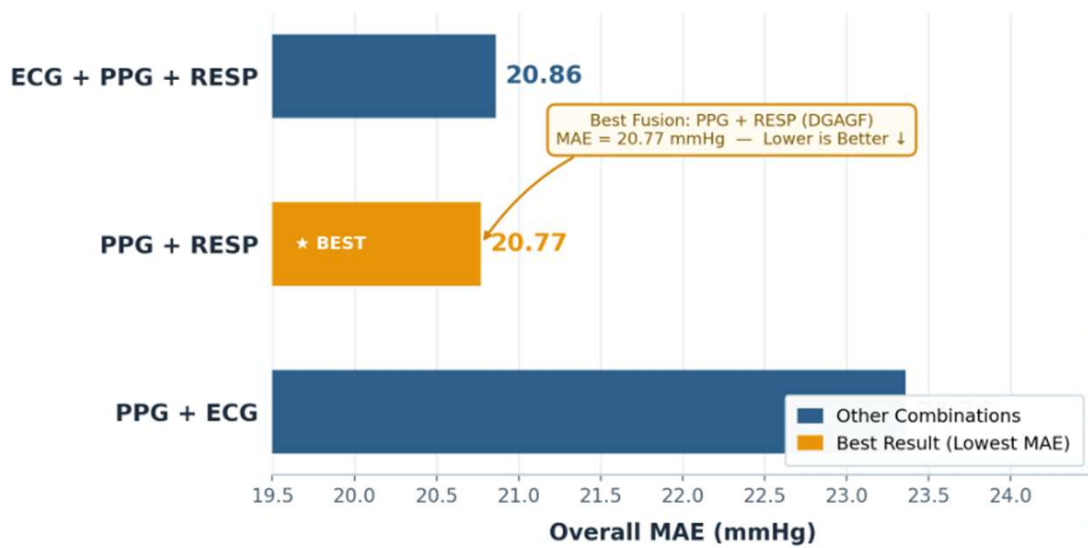


Experimental Results (2/2)

Multimodal Fusion Benchmark

Fusion Combination	Best Fusion Method	Overall MAE
PPG + ECG	DGAGF	23.36
PPG + RESP	DGAGF	20.77
ECG + PPG + RESP	DGAGF	20.86

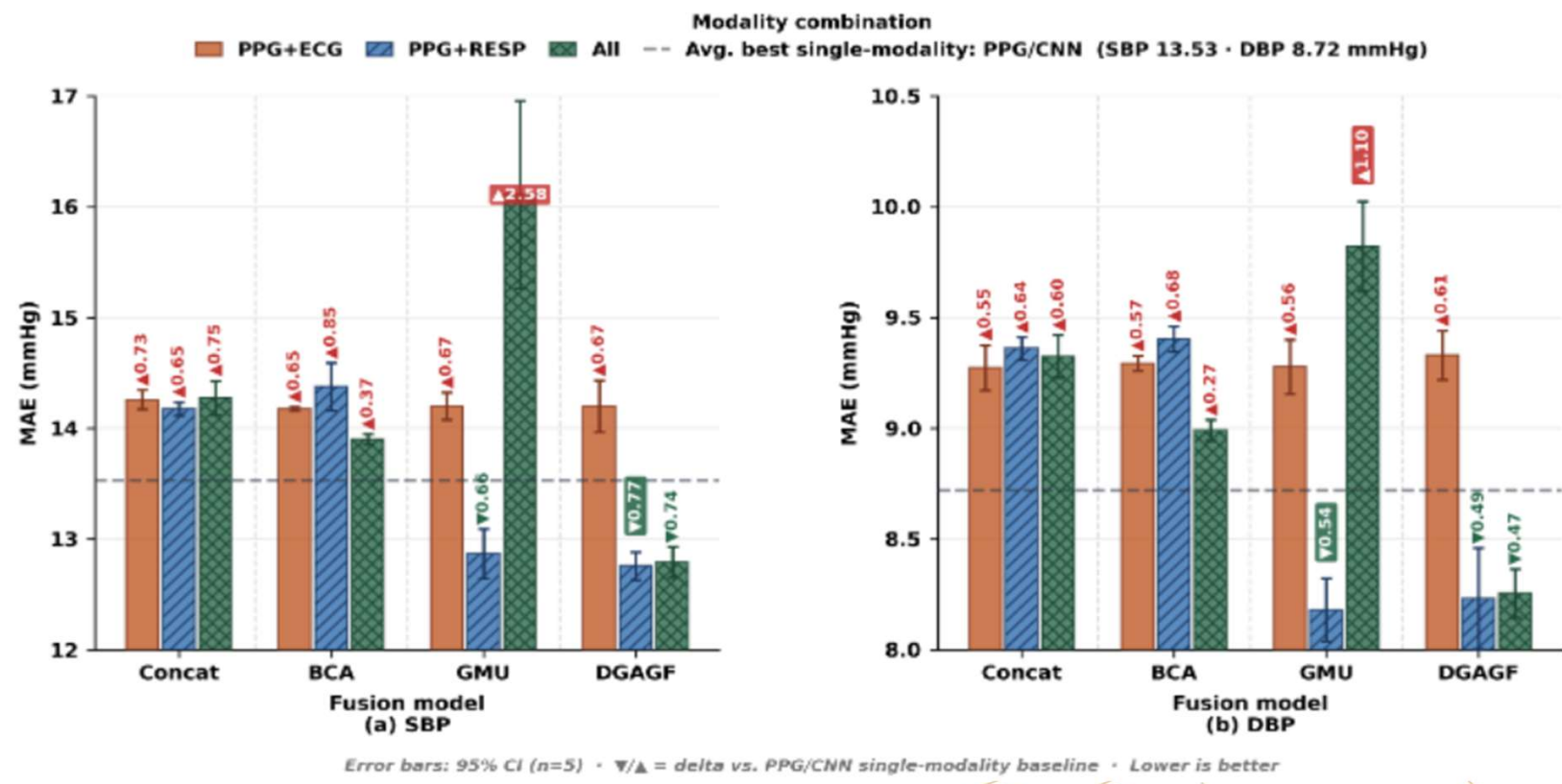
- DGAGF achieved the lowest overall MAE.
- PPG + RESP produced the lowest error.
- PPG + RESP outperformed other fusion combinations.
- More modalities did not always lead to better results.



Best Fusion Configuration
Fusion: PPG + RESP
Model: DGAGF
Overall MAE: 20.77

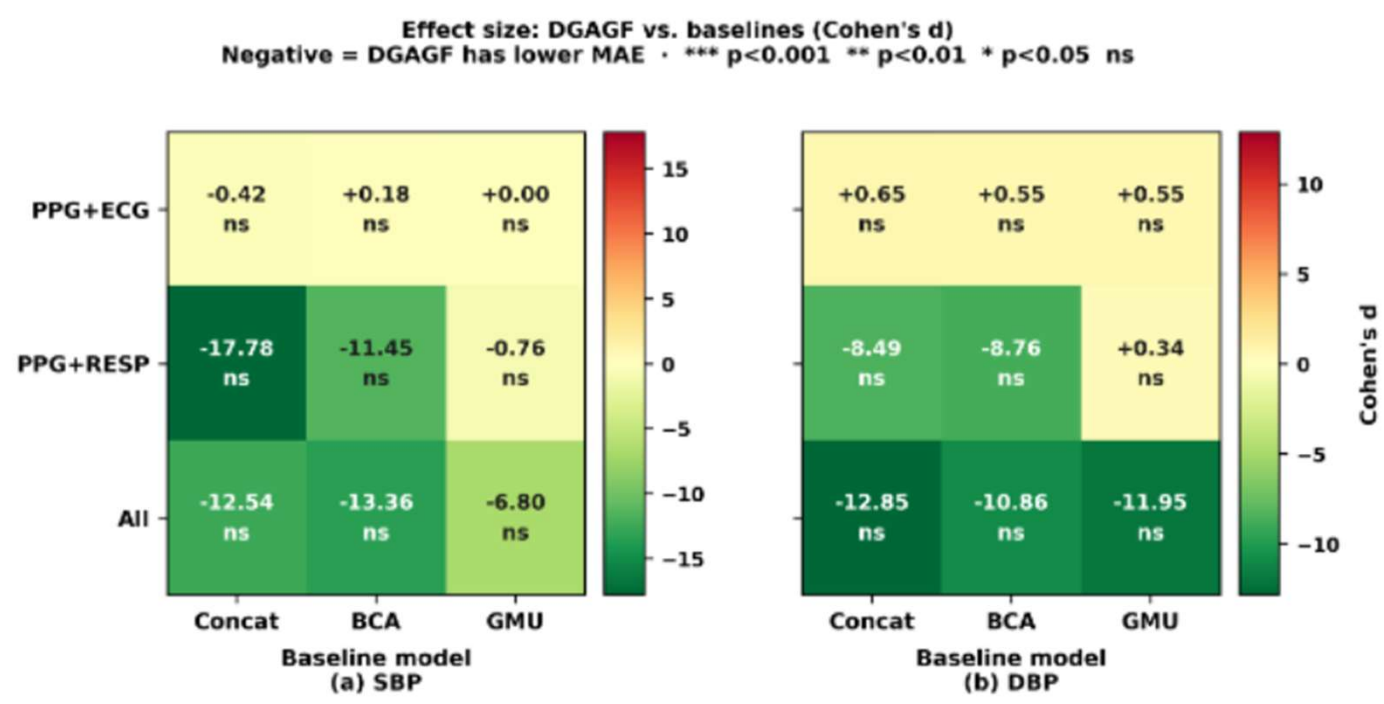
Experimental Results (2/2)

Multimodal Fusion Benchmark



Experimental Results (2/2)

Multimodal Fusion Benchmark



Study 2 findings



- DGAGF achieved the best fusion performance.
- Fusion strategy is more important than simply adding modalities.
- PPG + RESP was the best modality combination.
- Dual-modality outperformed tri-modality.
- DGAGF reduced the impact of noisy modalities.
- DGAGF provided more stable predictions.

Application Interface

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AR * ≡

Patient

Relative

Sign In

Sign Up

Patient Sign In

PATIENT ID

e.g. P001

PASSWORD

Sign In

 SENTINEL

Home Patient Portal Relative Portal

AR *

Patient

Relative

Sign In

Sign Up

Patient Sign In

PATIENT ID

e.g. P001

PASSWORD

Sign In



SENTINEL

AI-Powered Blood Pressure Monitoring

ECG
SIGNAL

PPG
SIGNAL

AI
ANALYSIS



Dashboard Demo



SENTINEL

Home

Patient Portal

Relative Portal

AR



I'm a Patient

I'm a Relative



AI-Powered Analysis

Multi-signal ECG, PPG & respiratory analysis using deep learning.



Real-Time Alerts

Instant notifications to family members when readings are critical.



Elderly-Friendly

Designed for simplicity — large text, clear icons, one-tap operation.



SENTINEL — Advanced Healthcare Monitoring

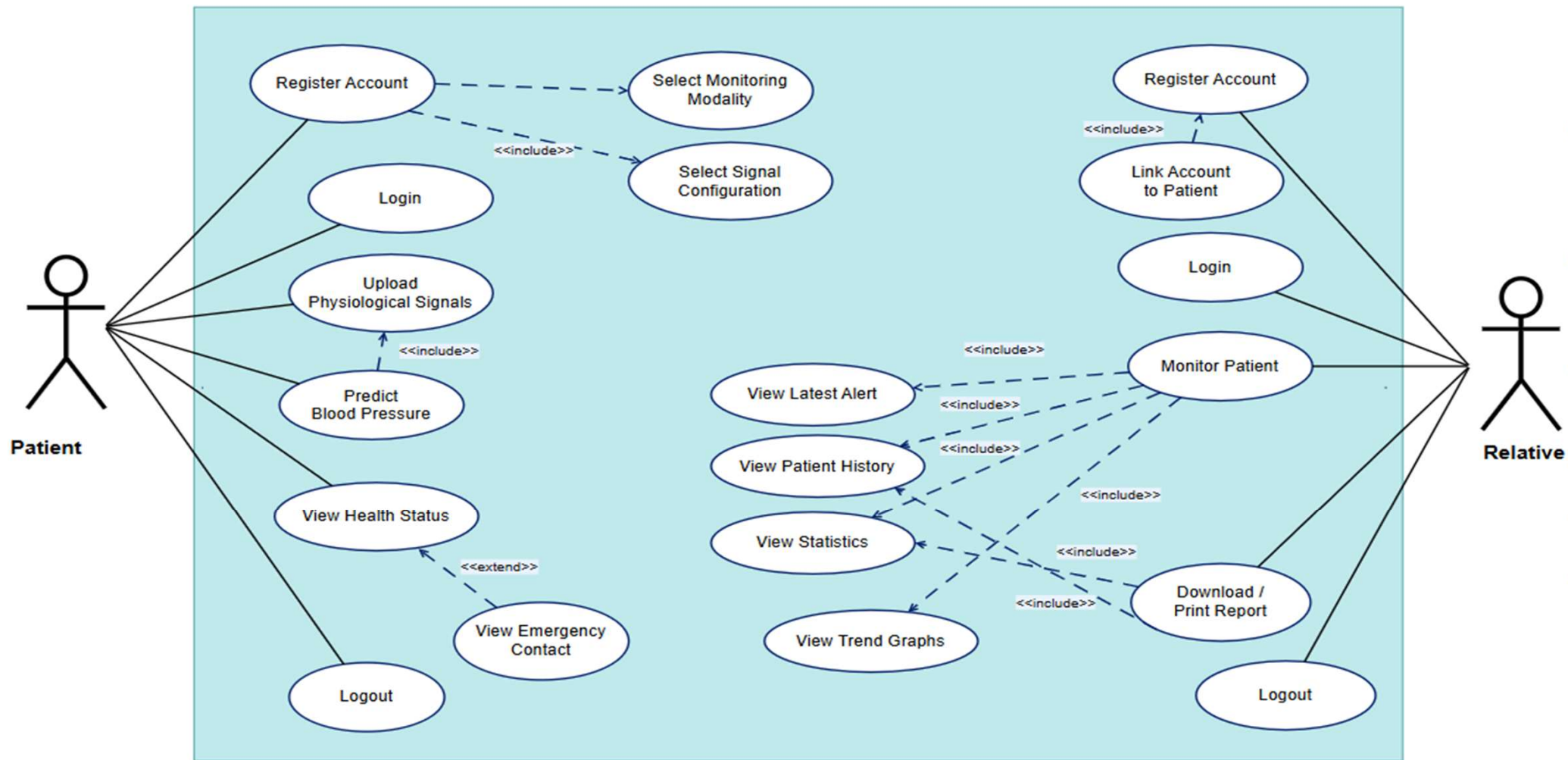
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Tools & Technologies

Component	Technology
AI Model Development	Python, TensorFlow, Keras
Frontend	React
Backend	FastAPI
Database	SQLite
Testing	Postman, Black-box Testing
Development Environment	VS Code, Kaggle, Google Colab

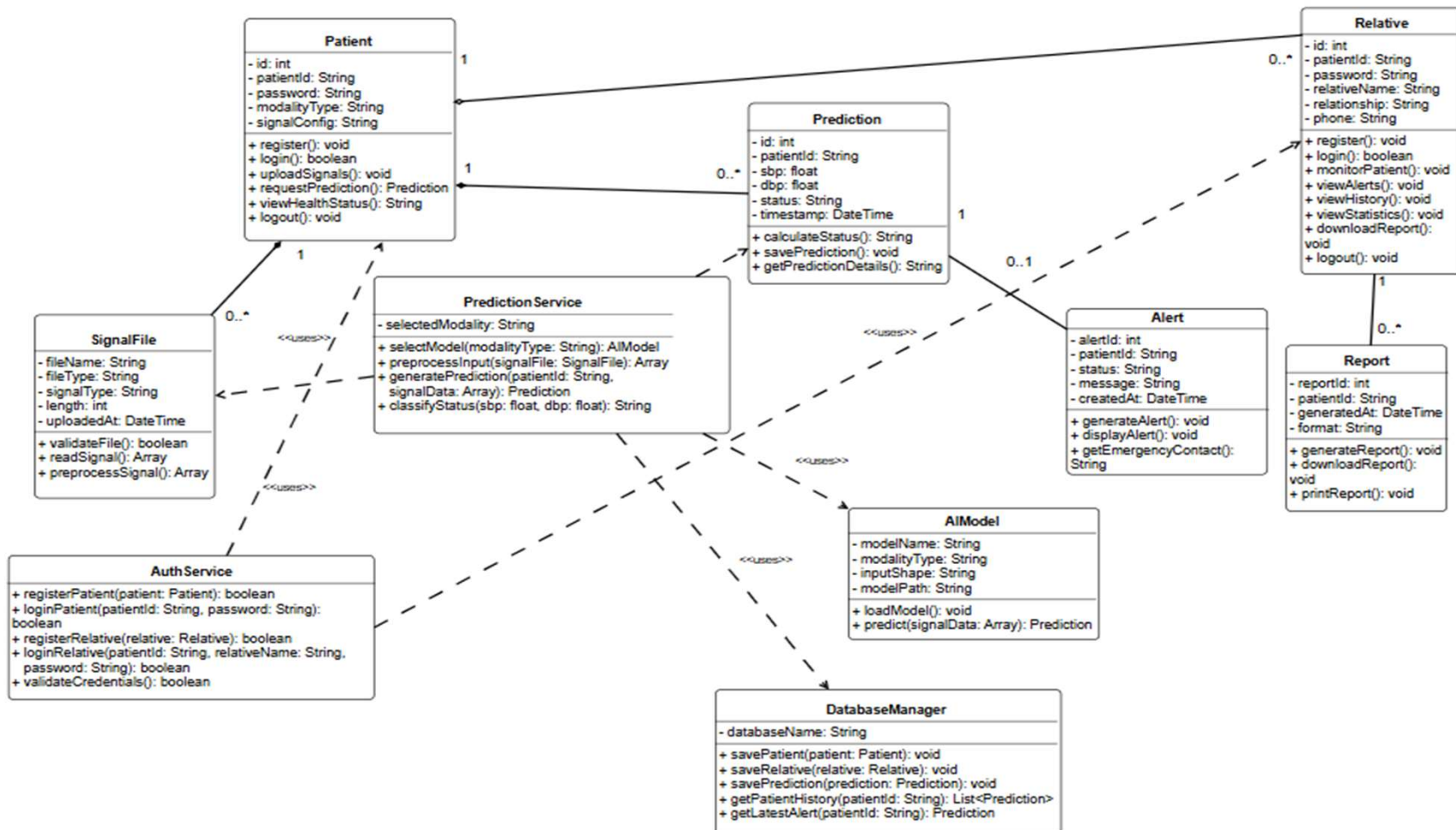
Diagrams

Use Case Diagram



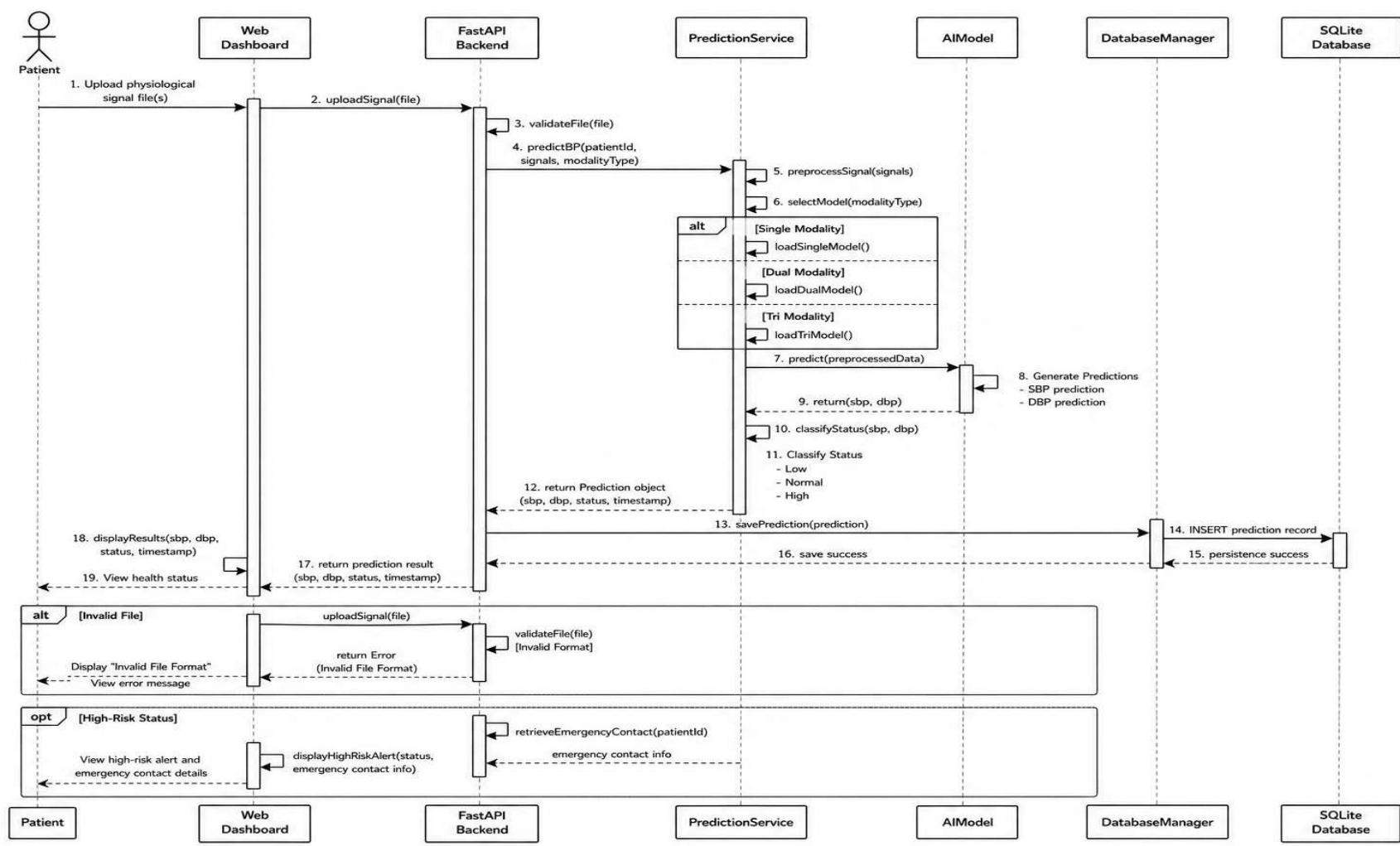
Diagrams

Class Diagram



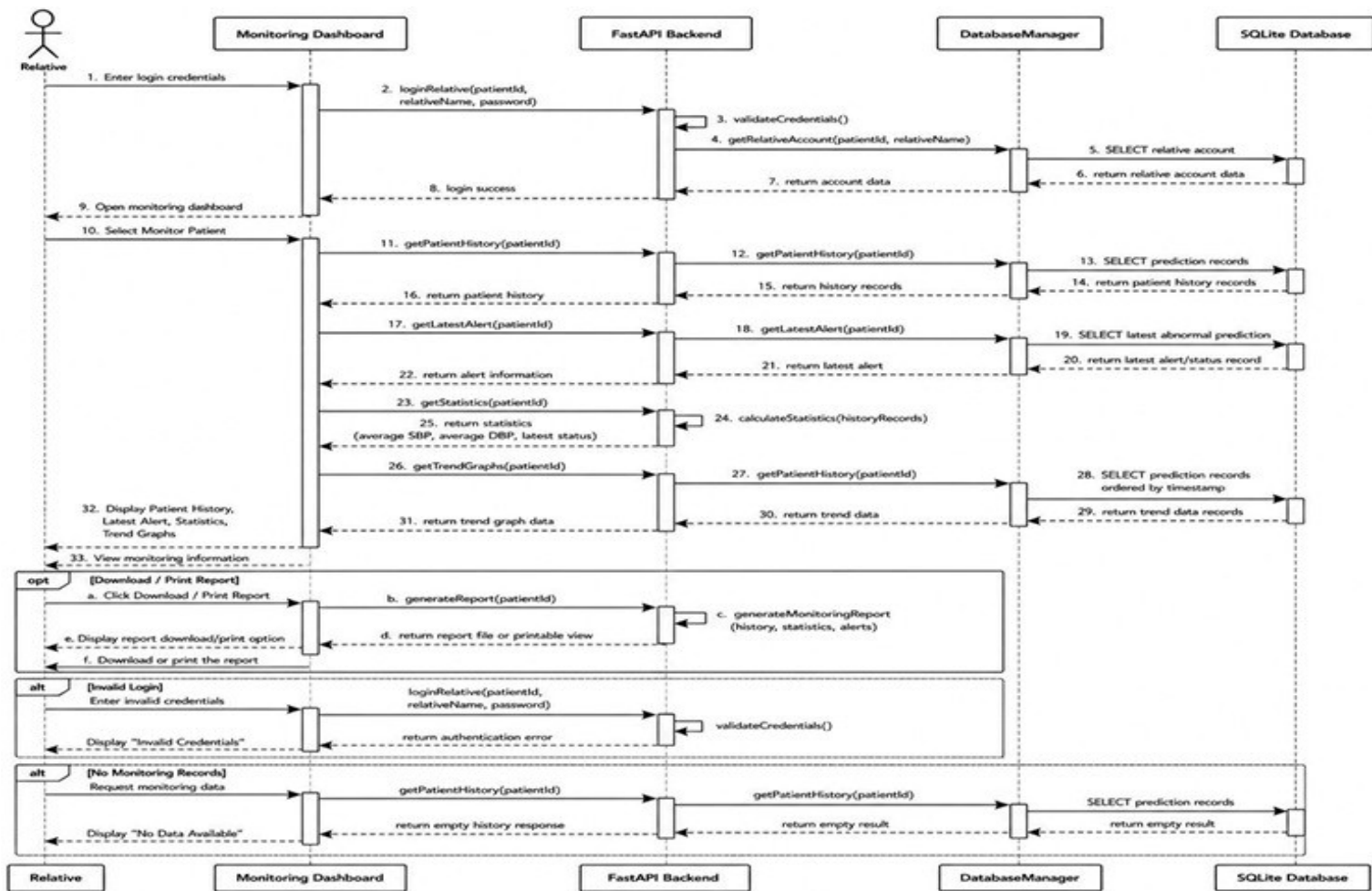
Diagrams

Sequence Diagram



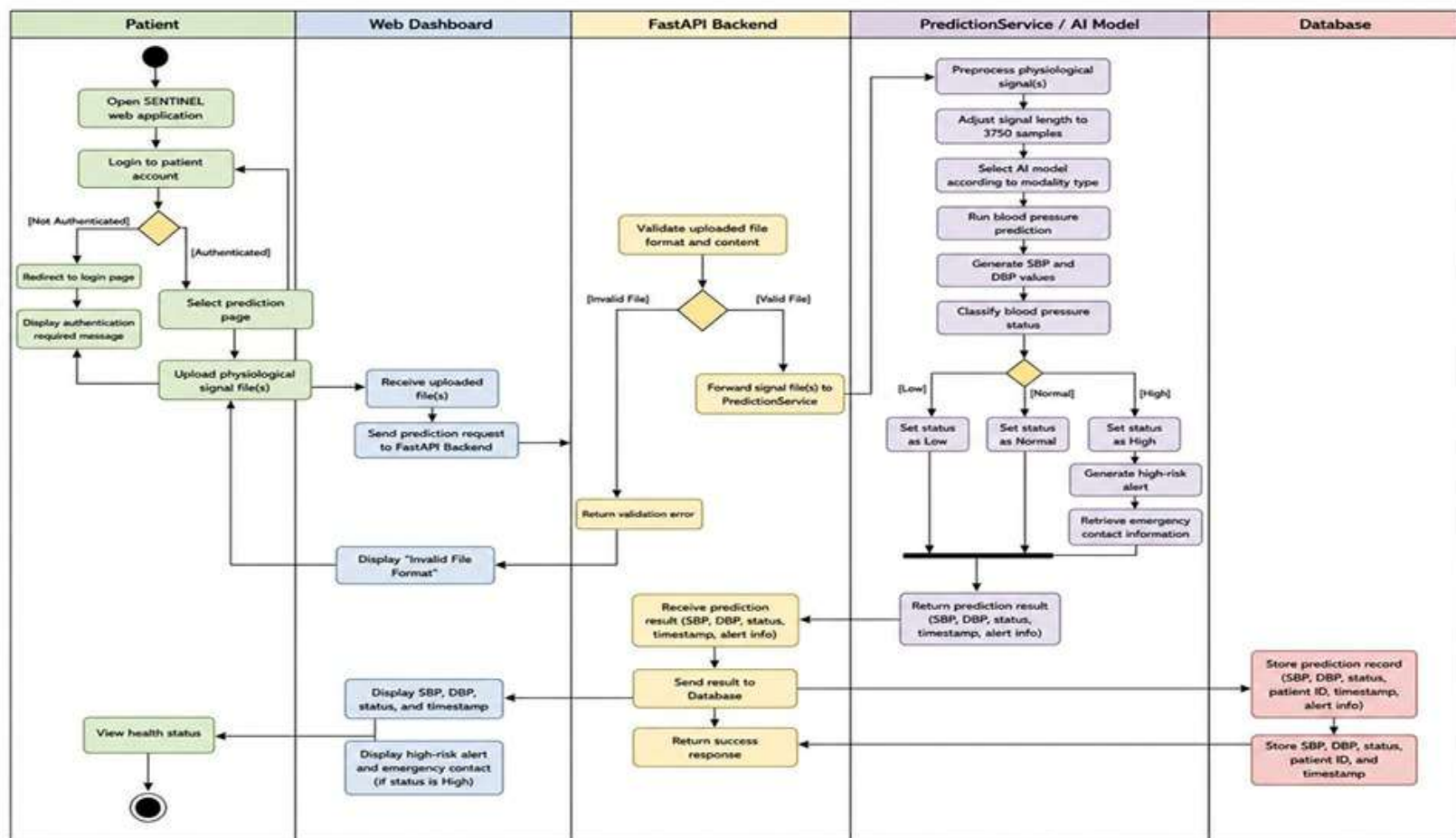
Diagrams

Sequence Diagram



Diagrams

Activity Diagram



Challenges & mitigation

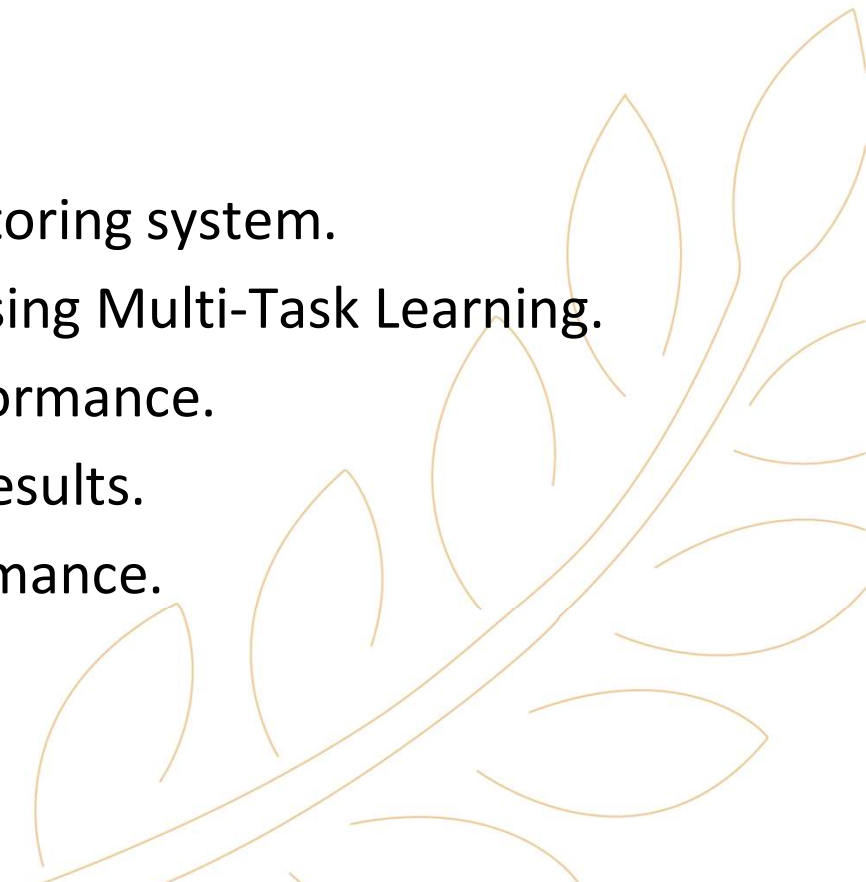
Challenge	Mitigation
Data Leakage	Subject-Level Split
Fair Modality Comparison	Unified preprocessing pipeline for all modalities
Fusion Strategy Selection	Comparison of Concatenation, BCA, GMU, and DGAGF
Unequal Information Contribution Across Modalities	Proposed DGAGF dominant-guided fusion
AI Model Integration	FastAPI Backend Integration
Historical Monitoring	SQLite Prediction Database
Patient Monitoring by Relatives	Secure Patient-Relative Linking

Contributions

Contribution	Description
Fair Modality Benchmark	Compared ECG, PPG, and RESP under the same dataset and preprocessing pipeline
Architecture Comparison	Evaluated CNN, RNN, LSTM, GRU, and Bi-LSTM models
Multi-Task Prediction	Predicted SBP and DBP simultaneously using one model
Multimodal Fusion Analysis	Compared ECG+PPG, PPG+RESP, and ECG+PPG+RESP combinations
DGAGF Framework	Proposed dominant-guided fusion using PPG as the main modality
AI-Powered Monitoring Platform	Integrated prediction models into a dashboard for patients and relatives

Conclusion

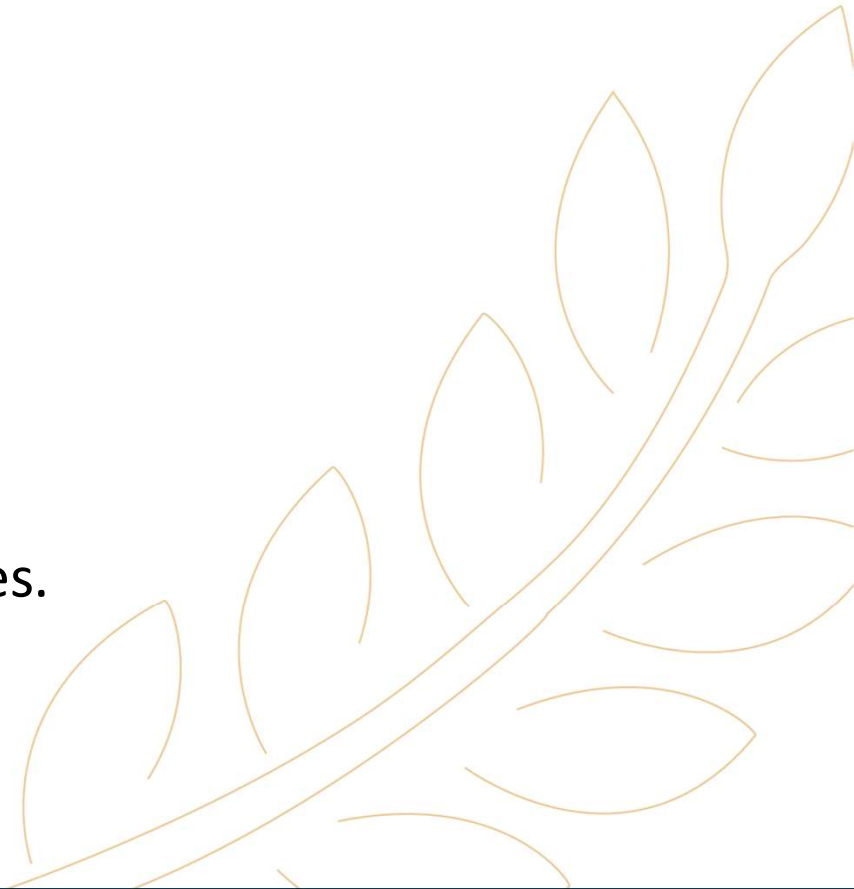


- Developed an AI-powered cuffless BP monitoring system.
 - Successfully estimated both SBP and DBP using Multi-Task Learning.
 - PPG achieved the best single-modality performance.
 - PPG + RESP achieved the best multimodal results.
 - DGAGF improved multimodal fusion performance.
- 

Future Work



- Real-time wearable integration.
- Mobile application deployment.
- Edge AI optimization.
- Clinical validation with human expert.
- Integration with commercial wearable devices.



Research Outcomes

- Paper 1 (Accepted)
- Accepted at [DATA'26]



Cuffless Blood Pressure Monitoring via Multi-Task Learning of Physiological Time-Series. DATA 2026. Rome, Italy [Accepted]

- Paper 2 (Under Review)
- Under Review at [ARTIIS 2026]



Dear authors,

We received your submission to ARTIIS 2026 (Conference on Advanced Research in Technologies, Information, Innovation and Sustainability):

- Authors: Rawan A. Ahmed , Menna Hossameldin Mokdam , Moamen Zaher , Mohamed N. Saad , Tamer M. Nassef
- Title: Multimodal Cuffless Blood Pressure Estimation Using Gated Fusion and Multi-Task Learning
- Number: 6275
- Track: ARTIIS 2026 - Main Conference

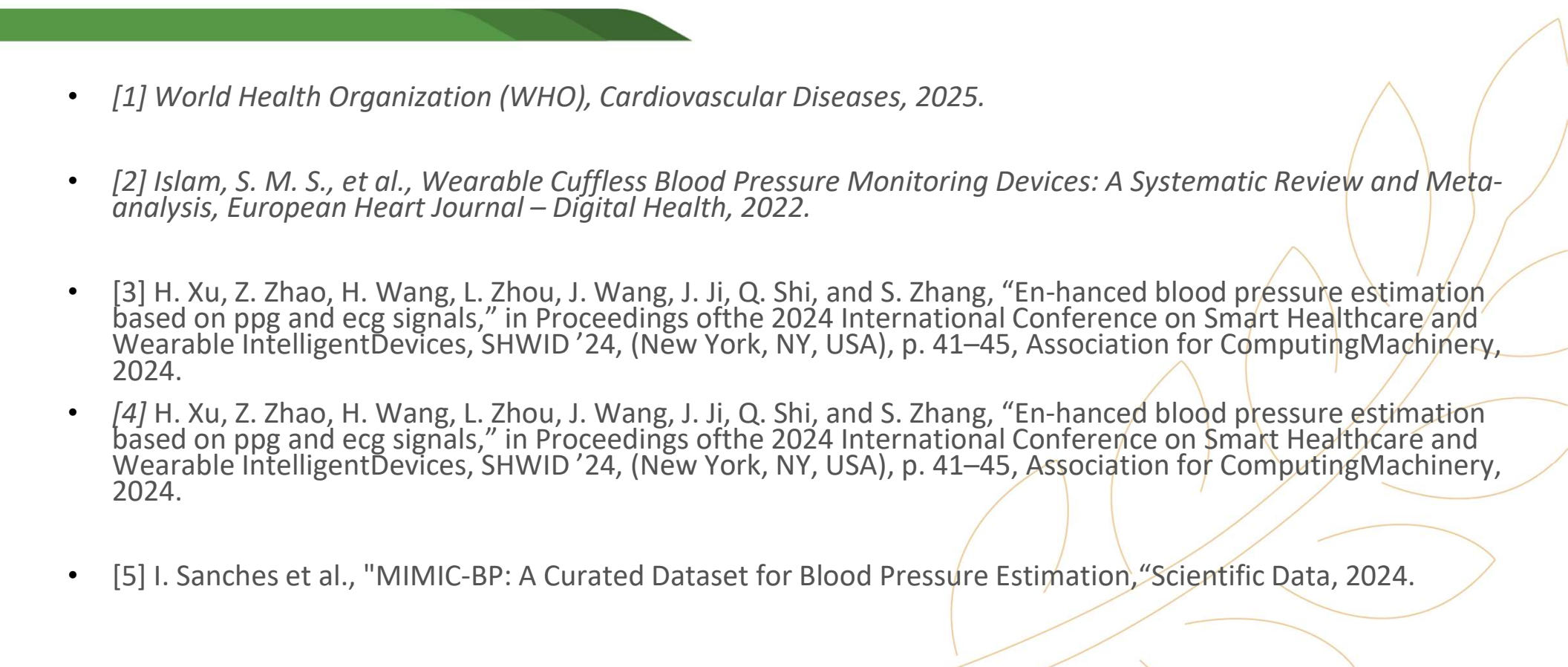
The submission was uploaded by Rawan A. Ahmed . You can access it via the [ARTIIS 2026 EasyChair Web page](#).

Thank you for submitting to ARTIIS 2026.

Best regards,
EasyChair

Multimodal Cuffless Blood Pressure Estimation Using Gated Fusion and Multi-Task Learning. ARTIIS 2026. Lisbon, Portugal [Under Review]

References



- [1] World Health Organization (WHO), *Cardiovascular Diseases*, 2025.
- [2] Islam, S. M. S., et al., *Wearable Cuffless Blood Pressure Monitoring Devices: A Systematic Review and Meta-analysis*, *European Heart Journal – Digital Health*, 2022.
- [3] H. Xu, Z. Zhao, H. Wang, L. Zhou, J. Wang, J. Ji, Q. Shi, and S. Zhang, “En-hanced blood pressure estimation based on ppg and ecg signals,” in *Proceedings of the 2024 International Conference on Smart Healthcare and Wearable Intelligent Devices, SHWID '24*, (New York, NY, USA), p. 41–45, Association for Computing Machinery, 2024.
- [4] H. Xu, Z. Zhao, H. Wang, L. Zhou, J. Wang, J. Ji, Q. Shi, and S. Zhang, “En-hanced blood pressure estimation based on ppg and ecg signals,” in *Proceedings of the 2024 International Conference on Smart Healthcare and Wearable Intelligent Devices, SHWID '24*, (New York, NY, USA), p. 41–45, Association for Computing Machinery, 2024.
- [5] I. Sanches et al., “MIMIC-BP: A Curated Dataset for Blood Pressure Estimation,” *Scientific Data*, 2024.



Any Questions?

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